

LEXORICH-M

Levocetirizine Dihydrochloride 5 mg + Montelukast Sodium 10 mg Tablets

1. Pharmacodynamics

LEXORICH-M combines a selective second-generation antihistamine with a leukotriene receptor antagonist to provide dual-pathway control of allergic inflammation.

Levocetirizine Dihydrochloride

Levocetirizine is the active R-enantiomer of cetirizine and acts as a potent, selective peripheral H1-receptor antagonist.

Its pharmacological actions include:

- Inhibition of histamine-mediated allergic responses
- Reduction of sneezing, rhinorrhea, nasal itching, and ocular symptoms
- Decrease in eosinophil migration and inflammatory mediator release
- Minimal penetration across the blood-brain barrier compared with first-generation antihistamines

Levocetirizine provides rapid symptomatic relief in allergic conditions while maintaining a relatively low sedative profile.

Montelukast Sodium

Montelukast is a selective cysteinyl leukotriene receptor (CysLT1) antagonist that inhibits the action of leukotrienes LTC₄, LTD₄, and LTE₄.

Leukotrienes are important mediators involved in:

- Airway edema
- Bronchoconstriction
- Mucus secretion
- Nasal congestion
- Eosinophilic airway inflammation

By blocking leukotriene receptors, Montelukast improves both upper and lower airway inflammatory symptoms associated with allergic rhinitis and asthma.

Dual Mechanism of Action

The Levocetirizine-Montelukast combination targets two major inflammatory pathways in allergic disease:

- Histamine-mediated immediate allergic responses
- Leukotriene-mediated persistent inflammatory symptoms

This dual-action approach may provide broader symptom control compared with antihistamine monotherapy, particularly in patients with nasal congestion, nocturnal symptoms, and concomitant airway hyperreactivity.

2. Indications

LEXORICH-M is indicated for the management of:

- Seasonal allergic rhinitis
 - Perennial allergic rhinitis
 - Allergic rhinoconjunctivitis
 - Allergic symptoms associated with bronchial asthma
 - Patients with allergic rhinitis and comorbid asthma
 - Persistent nasal congestion associated with allergic inflammation
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3. Research-Backed Benefits

Clinical investigations evaluating antihistamine-leukotriene antagonist combinations have demonstrated improved control of multiple allergic symptoms compared with antihistamine therapy alone.

Night-Time Symptom Control

Studies involving Montelukast and Levocetirizine combinations have shown improved reduction in:

- Night-time nasal obstruction
- Sleep disturbance associated with allergic rhinitis
- Nocturnal coughing related to allergic airway inflammation

Leukotriene-mediated inflammation is believed to contribute significantly to persistent night-time symptoms, which may not be adequately controlled with isolated H1-antagonism.

Nasal Congestion

Conventional antihistamines are often less effective against nasal congestion because congestion is mediated not only by histamine but also by leukotrienes and vascular inflammatory pathways.

Montelukast supplementation may provide:

- Improved nasal airflow
- Reduction in mucosal edema
- Better control of persistent congestion symptoms

Clinical observations suggest that the combination may offer superior relief in patients with moderate-to-severe allergic rhinitis, particularly those with inflammatory airway involvement.

Patients with Allergic Rhinitis and Asthma

The combination may be beneficial in individuals with overlapping upper and lower airway allergic disease due to Montelukast's effects on bronchial leukotriene-mediated inflammation.

4. Dosage and Administration

Adults and Adolescents (≥15 years)

- One tablet once daily or as directed by the physician.

Method of Administration

- Administer orally with water.
- May be taken with or without food.
- Evening administration is often preferred in patients with prominent nocturnal symptoms.

5. Special Warnings and Precautions

Neuropsychiatric Events

Serious neuropsychiatric events have been reported with Montelukast, including:

- Agitation
- Anxiety
- Sleep disturbances
- Depression
- Hallucinations
- Suicidal ideation and behavior (rare)

Patients and caregivers should be advised to monitor for behavioral or mood changes. Discontinuation should be considered if neuropsychiatric symptoms occur.

Renal Impairment

Levocetirizine is substantially excreted via the kidneys.

Use with caution in:

- Moderate renal impairment
- Severe renal dysfunction

Dose adjustment may be required based on creatinine clearance.

Hepatic Impairment

Montelukast should be used cautiously in severe hepatic dysfunction.

Sedation

Although Levocetirizine is considered a low-sedating antihistamine, somnolence and fatigue may occur in susceptible individuals.

Patients should be cautioned regarding activities requiring mental alertness if drowsiness develops.

6. Drug Interactions

Alcohol and CNS Depressants

Concurrent use with alcohol or central nervous system depressants may enhance sedation and psychomotor impairment in sensitive individuals.

Theophylline

Co-administration with theophylline may modestly reduce Montelukast plasma clearance, although this is generally not considered clinically significant.

Phenobarbital and CYP Enzyme Inducers

Enzyme inducers such as phenobarbital, rifampicin, and phenytoin may reduce Montelukast plasma concentrations.

Other Antihistamines

Concurrent use with additional antihistamines may increase the risk of additive adverse effects including drowsiness.

7. Contraindications

LEXORICH-M is contraindicated in patients with:

- Known hypersensitivity to Levocetirizine, cetirizine, Montelukast, or formulation components
 - Severe renal impairment in the absence of appropriate dose adjustment
 - History of hypersensitivity reactions to piperazine derivatives
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8. Adverse Reactions

Common Adverse Effects

- Headache
- Somnolence
- Fatigue
- Dry mouth
- Dizziness

Gastrointestinal Effects

- Abdominal pain
- Dyspepsia
- Nausea

Neuropsychiatric Effects

- Irritability
- Sleep disturbances
- Anxiety (rare)

Hypersensitivity Reactions

- Rash
- Pruritus
- Angioedema (rare)

Most adverse effects are mild to moderate in severity and transient in nature.

9. Storage

Store below 25°C in a dry place, protected from moisture and direct sunlight.
Keep out of reach of children.

10. Dosage Form

Film-coated tablets for oral administration.